

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

ATTACHMENT E

NARRATIVES OF INVESTIGATIVE INTERVIEWS

Ironworker Foreman

Laborer #1

Laborer #2

Ironworker

(10 Pages)



**NATIONAL TRANSPORTATION SAFETY BOARD
WASHINGTON, DC 20594**

**NARRATIVE OF INVESTIGATIVE INTERVIEW
CONCERNING THE ACCIDENT WHICH OCCURRED ON JULY 10, 2006
BOSTON, MASSACHUSETTS
HWY-06-MH-024**

Interviewee:

Mr. John Forsyth

Interviewed by:

Robert Accetta¹

Dennis Collins

Others present:

None

The interview was conducted at Mr. Forsyth's residence on November 8, 2006, and began at 1050 hours local time.

According to Mr. Forsyth, at the time he worked on in the I-90 Connector Tunnel, he had approximately 20 years of construction experience. He stated he only worked on this project for two weeks in 1999. While in the Tunnel, he was employed by Modern Continental Construction (MCC) as a Foreman. As a Foreman, he was in charge of a work crew consisting of five ironworkers. During his two weeks in the Tunnel, his crew consisted of the same persons.

Mr. Forsyth stated that each morning he would meet with the general foreman at a "tool box meeting" for his work assignment. His crew was assigned to install the 45-degree angle tie rods to the ceiling and panel and ensure the panel was plumb. Mr. Forsyth stated he did not observe the installation of the bolts or any testing conducted to the bolts before or after module installation.

According to Mr. Forsyth, when he and his crew reported to the panel, it had already been suspended from the ceiling by the vertical tie rods (work which had been performed by another work crew). Mr. Forsyth stated that occasionally the panel might also have been suspended from a 45-degree angle tie rod for horizontal (left/right) adjustment. Mr. Forsyth stated that each module had been plumbed as close as possible to the required standards before his crew completed the tie rod installation. He further stated that the panels could have been suspended by the verticals alone for up to a week

¹ National Transportation Safety Board (NTSB)

before the 45-degree angle tie rods were installed, and that more than one panel would be suspended, waiting for 45-degree rods, at one time. Mr. Forsyth believes the panels were hung from all the vertical supports for that panel.

Mr. Forsyth estimated it would take the crew 10 to 15 minutes to complete the installation of two 45-degree angle tie rods and set them. While on the module, one part of the crew would set out the hardware while the other part of the crew would put the pins in place and set the hardware. Mr. Forsyth compared the process to an assembly line. He could not recall if the required hardware was laid out on the module before the crew's arrival or if the crew had to bring it up, but he believes the hardware had been laid out on the module by the previous crew. Mr. Forsyth also stated he does not recall having to disconnect any of the vertical rods to connect the 45-degree rods.

When asked to describe the procedure used to plumb the module, Mr. Forsyth stated that the panels were very close to their final position when they were set on the verticals, indicating they typically only needed to be adjusted from 1/8 to 1/4 of an inch. Mr. Forsyth stated that for left-to-right leveling, they would use elevation marks previously placed on the walls to run a "dry string" from one point to another. When asked to explain a "dry string", Mr. Forsyth stated the crew would run a string with a 1/4 inch shim and set the panel using that shim. A transit (level) would also be used to match the ends of the panel with the adjacent (already placed) panels.

When asked, Mr. Forsyth stated he believes the engineers providing direction on the leveling of the panel were in fact up on the panel during the leveling operation.

When asked if anything unusual about the tunnel stuck out in his mind, Mr. Forsyth stated neither he nor the crew liked the epoxy and bolt system, calling it "shabby". He stated he had never seen panels of this weight suspended from the ceiling using epoxy prior to working in the tunnel. Since his work with Modern Continental, Mr. Forsyth has installed aluminum panels for Tower Glass on the Silver Bus Line. These panels used a channel connection where the bolts were shot directly into the concrete ceiling. These aluminum panels were being used for cosmetic (architectural) purposes only and were much lighter than the concrete panels.

When asked if he was concerned working up on the panels, Mr. Forsyth stated he was not, as the safety rules required the workers to wear a harness, glasses, vest, and hardhat, as well as be tied off. According to Mr. Forsyth, Safety Inspectors employed by Modern Continental would monitor the workers. He did not recall what lighting they had on top of the panel, but believes some sort of lighting was provided.

The interview concluded at 1142 hours.



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Interviewed by:

Robert Accetta²
Dennis Collins

Others present:

David Vicinanzo³
D. Danielle Pelot
Eric F. Eisenberg⁴
Pedro Pinheiro⁵

The interview was conducted at the offices of Nixon Peabody LLP, 100 Summer Street, Boston, MA 02110-2131 on November 9, 2006.

According to Mr. Pereira, he began in construction, joining the Union, in 1994. At the time he worked in the I-90 Connector Tunnel, Modern Continental Construction employed him as a Laborer. He stated it was his job to drill the anchor holes, clean the holes, and insert the epoxy and bolts into the holes. According to Mr. Pereira, he was not involved in hanging the panels; ironworkers handled that job.

When asked to describe the drilling process, Mr. Pereira stated that he and an assistant comprised a two-man team, and they were the only team drilling holes. Mr. Pereira would operate the drill and his assistant would operate the water used in the drilling process. The two men would use a scissors-lift to raise themselves, the drill, and their other equipment to the ceiling of the tunnel. Both men would raise the drill to the ceiling, lined up with a mark on the ceiling, and secured to the ceiling using a vacuum attachment. Mr. Pereira would slowly raise the bit, using a small crank on the drill stand, ensuring the bit was squarely on the mark. Mr. Pereira stated surveyors had placed the mark on the ceiling of the tunnel. When the hole had been drilled to the correct depth, Mr. Pereira and his assistant would crank the bit down, disconnect the drill from the ceiling, clean the concrete core out of the drill bit, and check the depth of the hole with a tape measure. Mr. Pereira estimated it would take between 5 and 10 minutes to drill a

² National Transportation Safety Board (NTSB)

³ Nixon Peabody LLP

⁴ Hinckley, Allen, Snyder LLP

⁵ American Translation Partners

hole in the ceiling; this would increase by more than 5 minutes if they need to drill through rebar. According to Mr. Pereira, the scissors-lift was 6 to 8 feet long and would allow them to drill 6 to 8 holes before they would need to lower and reposition the lift. Mr. Pereira stated they would drill holes for several days or a week before proceeding to the next step, cleaning the holes.

When asked how many holes he could drill a day, Mr. Pereira stated he was not sure, but would guess that he could do between thirty (30) and forty (40) holes a day, being able to drill between twenty (20) and twenty-five (25) holes before needing a new bit. He was not sure if he needed to change the drill bit when drilling through rebar. When asked how he was sure he had drilled to the correct depth, Mr. Pereira stated both the drill bit and the drill stand crank were marked. He would mark any new bits to indicate the correct depth. When asked if he recalled the drill speed, he stated he believed the drill only had one speed. He did not recall having any problems while drilling. Mr. Pereira also stated he would check the depth of the drilled hole with a tape measure. According to Mr. Pereira, he would sometimes have to use a long screwdriver to remove a portion of the core still in the hole.

According to Mr. Pereira, cleaning the holes involved using an “iron” (round wire) brush six (6) to seven (7) inches long. The brush was inserted in the hole and moved along its length several times. Next, Mr. Pereira used compressed air to blow out the holes. Once the holes were clean, they would be allowed to dry for an additional two (2) to three (3) days.

Inserting the bolts was again described as a two-man procedure requiring the use of a scissors-lift. One member of the team would inject the epoxy and the other would insert the bolts. According to Mr. Pereira, they could get to between six (6) and eight (8) holes before having to move the lift. When injecting the epoxy into the holes, Mr. Pereira would insert a “red cap” into the hole. He would insert the nozzle of the epoxy through a slit in the red cap and into the hole. Mr. Pereira would bottom out the nozzle in the hole and slowly withdraw the nozzle while squeezing the trigger of the epoxy gun. He stated he would fill the hole with epoxy. The gun used to dispense the epoxy was the manual version. Mr. Pereira stated that when he started using a new tube of epoxy, he would run some of the material out as waste to ensure he was placing properly mixed epoxy in the holes. He believes the tips of the nozzles were at a 45-degree angle, but he did not cut them; the nozzles came that way. When asked, Mr. Pereira indicated through the interpreter that he believed the epoxy used was the fast-set version, but he was not sure. He does not recall checking the expiration date on the epoxy when he opened a new tube. Mr. Pereira stated that leaving the epoxy still for five minutes would cause it to harden in the nozzle, forcing him to replace the nozzle. He also stated that on some occasions, moving the lift took more than five minutes, resulting in a nozzle replacement.

After the hole had been filled with epoxy, Mr. Pereira would insert a pre-cut anchor (bolt) into the hole through the red cap. He stated that he could feel resistance from the epoxy when he did this. He stated he would insert the bolt through the red cap, and then twist it as he pushed it up into the hole, stopping when the bolt reached the

bottom of the hole. The twist was described as being completely around, a full 360-degree twist. Mr. Pereira stated he used rags to clean up epoxy that was forced from the hole as he inserted the bolt. He then measured the portion of the bolt remaining outside the hole to be sure it was the correct length.

Mr. Pereira stated he did have to re-drill holes and re-insert epoxy after some of the bolts failed during testing. He stated the surveyors would make new holes for the entire plate and Mr. Pereira would drill them.

According to Mr. Pereira, his safety equipment consisted of a rain suit, goggles, gloves, and a hard hat. Personnel from Modern Continental or Bechtel/Parsons-Brinkerhoff would come up on the scissors-lift every other day or so to inspect his work.

When asked about training to drill the holes and use the epoxy, Mr. Pereira stated that both the manufacturer of the drill and the manufacturer of the epoxy provided training. The extent of the training was instruction, demonstration, and observation on the first day or so. Mr. Pereira stated he had a hard time with the first hole he drilled, but that it quickly got easier.



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Interviewee:

Mr. Duarte Caiano

Interviewed by:

Robert Accetta⁶
Dennis Collins

Others present:

David Vicinanza⁷
D. Danielle Pelot
Eric F. Eisenberg⁸
Pedro Pinheiro⁹

The interview was conducted at the offices of Nixon Peabody LLP, 100 Summer Street, Boston, MA 02110-2131 on November 9, 2006.

Mr. Caiano began his employment with Modern Continental Construction in 1999. During the course of his employment, he worked in the I-90 Connector Tunnel as a Laborer, his first job in construction. He described his duties as working with concrete, drilling holes, and cleaning up. He stated he worked as an assistant to Dimasio Pereira at first - for six (6) or seven (7) months - and then was placed in charge of the two-man crew on the lift and continued working on the ceiling for another one-and-a-half to two years. On the team, Mr. Caiano would operate the drill and his helper would operate the pump. While working as the lead on the lift, his helper could vary and he did not work on the ceiling every day. As far as Mr. Caiano knew, only two men were assigned to drill holes at any given time (either Pereira/Caiano or Caiano/Helper).

According to Mr. Caiano, the drilling process utilized a core drill with a water collection kit and a vacuum system to secure it to the ceiling of the tunnel. A diamond bit was used to allow them to drill through concrete and steel (rebar). The drill was mounted on a stand and the bit was raised to the ceiling with a small crank on the stand. Mr. Caiano stated he used the "regular" speed on the drill. He could not recall how deep the holes were to be drilled, but did remember the correct depth was marked on the drill

⁶ National Transportation Safety Board (NTSB)

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⁹ American Translation Partners

stand. He stated he checked the depth of the hole with a tape measure after he completed drilling. If some of the core remained in the hole, Mr. Caiano would break it off with a long screwdriver. He was unsure how many holes he could typically drill in a day. He also stated that an inspector, employed by Bechtel, checked the depth of the holes as well. Mr. Caiano noted that the drill had an auto-shutoff feature that would activate if the drill got wet. If the auto-shutoff feature activated, they would have to wait two (2) or three (3) days for the drill to dry before using it again. If the drill would not hold to the ceiling, the plate could be moved to allow drilling in a new location. When this was necessary, surveyors would move the plate and mark new hole locations to drill. Mr. Caiano stated he only drilled 4-hole plates –no 2-hole plates.

After drilling a hole, Mr. Caiano stated they would wait three (3) or four (4) days, and then clean the hole with a wire brush and compressed air generated by an electric compressor. Mr. Caiano could not recall the air pressure setting of the compressor. He stated that his help would occasionally assist by cleaning some of the holes. After the hole was cleaned, they would insert a “red cap” into the hole. According to Mr. Caiano, an inspector also looked at some of the holes after the cleaning was complete.

Next, Mr. Caiano and his helper would put epoxy and an anchor in the hole. According to Mr. Caiano, he would insert the nozzle through the “red cap” until it reached the bottom of the hole and then slowly withdraw the nozzle while squeezing out the epoxy using a manual gun. He stated he filled the hole with epoxy. He would then insert an anchor into the hole until it reached the bottom, giving it a complete 360-degree twist as he did so. He described the process of inserting the anchor as “fairly easy”. He would clean up any epoxy that came out of the hole using a rag and measure the length of the anchor remaining outside the hole. According to Mr. Caiano, sometimes the “red cap” would come out of the hole; he would simply re-insert it. Mr. Caiano stated he alone would fill the hole with epoxy and insert the anchor, not his helper. When asked, he stated he would run a little of the epoxy out when beginning a new tube. The initial epoxy from a new tube was a different color, indicating to Mr. Caiano that it was not mixed properly; this is why he would discard it. He also indicated that the epoxy would set up in a couple of minutes. Mr. Caiano stated that Dimasio Pereira had shown him how to use the epoxy, as well as the drill.

According to Mr. Caiano, he did not check the expiration dates on the epoxy as he opened new boxes of epoxy, but he recalls that Fernando (his Foreman) only ordered two (2) or three (3) cases at a time. Fernando would supply the epoxy to Mr. Caiano when it was needed.

When asked about testing performed on the anchors after the epoxy had set, Mr. Caiano stated that every bolt was checked and marked as acceptable or unacceptable. Mr. Caiano could only recall one bolt failing; when that happened, the surveyors were called and the entire plate was moved. The surveyors marked the new locations to drill.

Mr. Caiano stated his protective gear consisted on a vest, a hard hat, and rubber gloves.



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At the time he worked in the I-90 Connector Tunnel, Mr. Hurd was a Journeyman ironworker with Local #7 and had approximately 20 years of construction experience. He was not involved with the drilling or use of epoxy in the tunnel ceiling. His primary job while working in the tunnel was rod and turnbuckle assembly.

As Mr. Hurd described it, hundreds if not thousands of turnbuckles were used on the project. They would arrive disassembled and would need to be unloaded, moved to the location they were needed, and assembled. Unloading could take anywhere from a day to a week. Once the pieces were moved, Mr. Hurd and his crew would assemble the pieces and stockpile the turnbuckles for future use. Mr. Hurd described assembly as screwing two end pieces to a center piece.

According to Mr. Hurd, the installation of the turnbuckles involved loading the maximum amount of workers and equipment onto a scissors-lift and raising it to the ceiling. The plates would already be installed. Mr. Hurd stated he may have installed some plates, but cannot recall. The turnbuckles forming vertical supports were installed, then the crew pulled back and set the turnbuckles to the required elevation. Another

¹⁰ National Transportation Safety Board (NTSB)

¹¹ Nixon Peabody LLP

¹² Hinckley, Allen, Snyder LLP

crew, referred to as the “raising gang” would raise the panels on a lift referred to as a “woody”. Mr. Hurd stated that the panels were attached to all the vertical turnbuckles before the “woody” was lowered. He could recall no unusual alignment issues.

Mr. Hurd stated that he may periodically filled in for absent workers on other crews, and he may have gone back on top of a panel once it had been hung. He could recall no problems with the anchors, plates, turnbuckles, or panels; he was adamant that any such problems would have made an impression, as they would have been issues related to his personal safety. When asked, Mr. Hurd could not recall witnessing any pull testing of the anchors.